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OPERATING SYSTEM CONCEPTS LAB EXPERIMENTS

EXP – 5: SCHEDULING ALGORITHMS (Round Robin Scheduling)

user123@localhost:~/scheduling

```
[user123@localhost ~]$ ls
abc.sh  awk      Desktop   Downloads  ifelse.sh  memory  page  Pictures    Public   REC       sample.sh   scheduling  SFA  system_calls  Videos
ab.sh   bankers  Documents Folder     loops.sh   Music   pc    posneg.sh   qns.txt  sample.c  sample.txt  script      sh   Templates     while.sh
[user123@localhost ~]$ cd scheduling
[user123@localhost scheduling]$ ls
a.out  fcfs.c  priority.c  sjf.c
[user123@localhost scheduling]$ vi rr.c
```

user123@localhost:~/scheduling — /usr/bin/vim rr.c

```c
#include <stdio.h>
int main(){
	int n, tq, completed = 0, completionTime = 0;

	printf("Enter number of processes: \n");
	scanf("%d", &n);

	printf("Enter Time Quantum: \n");
	scanf("%d", &tq);

	int at[n], bt[n], rem[n], tat[n], wt[n];
	float avgtat = 0, avgwt = 0;
	// Input
	for (int i = 0; i < n; i++)
	{
		printf("Enter AT and BT for P%d: \n", i + 1);
		scanf("%d %d", &at[i], &bt[i]);
		rem[i] = bt[i]; // Remaining time = Burst time
	}
	// Round Robin Logic
	while (completed < n)
	{
		int idle = 1;
		for (int i = 0; i < n; i++)
		{
			if (at[i] <= completionTime && rem[i] > 0) //
			{
				idle = 0; //
				if (rem[i] > tq) //3,5 /1>2
				{
					completionTime += tq; //1+2 ==3
					rem[i] -= tq; //5-2=3
				} else {
					completionTime += rem[i]; //
					tat[i] = completionTime - at[i]; // Turnaround Time
					wt[i] = tat[i] - bt[i]; // Waiting Time
					rem[i] = 0;
					completed++;
				}
			}

		}
		if (idle){
			completionTime++; // If no process arrived yet//1
		}
	}

	// Calculate averages
	for (int i = 0; i < n; i++)
	{
		avgtat += tat[i];
		avgwt += wt[i];
	}

	printf("\nAverage Turnaround Time: %.2f\n", avgtat / n);

	printf("Average Waiting Time: %.2f\n", avgwt / n);

	return 0;
}
```

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```
[user123@localhost ~]$ ls
abc.sh  awk      Desktop    Downloads  ifelse.sh  memory  page  Pictures   Public   REC       sample.sh   scheduling  SFA  system_calls  Videos
ab.sh   bankers  Documents  Folder     loops.sh   Music   pc    posneg.sh  qns.txt  sample.c  sample.txt  script      sh   Templates     while.sh
[user123@localhost ~]$ cd scheduling
[user123@localhost scheduling]$ ls
a.out  fcfs.c  priority.c  sjf.c
[user123@localhost scheduling]$ vi rr.c
[user123@localhost scheduling]$ gcc rr.c
[user123@localhost scheduling]$ ./a.out
Enter number of processes:
4
Enter Time Quantum:
2
Enter AT and BT for P1:
0 5
Enter AT and BT for P2:
1 4
Enter AT and BT for P3:
2 2
Enter AT and BT for P4:
4 1

Average Turnaround Time: 7.25
Average Waiting Time: 4.25
```